

SPARQL

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1 Query 1

Natural language query

Get all the properties that can be applied to instances of the `Politician` class (<http://dbpedia.org/ontology/Politician>).

From this query, we want to find all the properties that appear in instances of the `Politician` class. In RDF, an instance is described using triples. In this case like: `?politician ?property ?value`. So we have to find the instances that belong to the `Politician` class and look for any property used to describe those instances.

SPARQL query

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT ?property
WHERE {
    # Find resources that are instances of the Politician class
    ?politician a dbo:Politician . #rdf:type --> a

    # Obtain every property used to describe those instances
    ?politician ?property ?value .
}
```

property
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/1999/02/22-rdf-syntax-ns#type

Figure 1: Properties used by instances of `Politician` class.

The result contains one row for every matching triple. Therefore, the same property can appear many times because different politicians, or even the same politician, may have triples using the same property.

To count the total number of property occurrences returned by the previous query, we use `COUNT(?property)`.

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT (COUNT(?property))
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .

    # Obtain every property used to describe them
    ?politician ?property ?value .
}
```

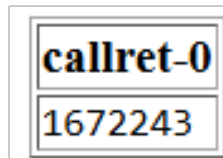


Figure 2: Total number of property occurrences for instances of `Politician` class.

If we only wanted to obtain each different property once (just to see which different properties there are), we can use `DISTINCT` in our query to remove the repeated properties.

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT DISTINCT ?property
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .

    # Obtain every property used to describe them
    ?politician ?property ?value .
}
```

property
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#comment
http://www.w3.org/2002/07/owl#sameAs
http://xmlns.com/foaf/0.1/name
http://xmlns.com/foaf/0.1/topic
http://www.w3.org/2007/05/powder-s#describedby
http://xmlns.com/foaf/0.1/homepage
http://purl.org/dc/terms/subject
http://xmlns.com/foaf/0.1/isPrimaryTopicOf
http://es.dbpedia.org/property/gabinete
http://es.dbpedia.org/property/afiliaciones
http://es.dbpedia.org/property/almaMáter
http://es.dbpedia.org/property/almaMáter
http://es.dbpedia.org/property/cargo
http://es.dbpedia.org/property/cargo

Figure 3: Different properties used by instances of `Politician` class.

And as done previously, we can use `COUNT(DISTINCT ?Property)` to count how many different properties there are.

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT COUNT(DISTINCT ?property)
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .

    # Obtain every property used to describe them
    ?politician ?property ?value .
}
```

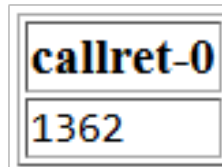


Figure 4: Number of distinct properties used by instances of Politician class.

2 Query 2

Natural language query

Get all the properties, except `rdf:type`, that can be applied to instances of the `Politician` class.

This query is similar to the previous one, but in this case we want to exclude the `rdf:type` property. To do this, we can use a `FILTER` condition that allows us to keep only the results where the property is different from `rdf:type`.

SPARQL query

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT ?property
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .

    # Obtain every property used to describe those instances
    ?politician ?property ?value .

    # Exclude rdf:type from the results
    FILTER (?property != rdf:type)
}
```

property
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#label

Figure 5: Properties used by instances of `Politician` class, excluding `rdf:type`.

The result contains one row for every matching triple, except those whose property is `rdf:type`. Therefore, repeated properties can still appear. We can use `COUNT` to see how many properties are after removing the `rdf:type` ones.

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT COUNT(?property)
WHERE {
```

```

# Find resources that are instances of Politician
?politician a dbo:Politician .

# Obtain every property used to describe those instances
?politician ?property ?value .

# Exclude rdf:type
FILTER (?property != rdf:type)

}

```

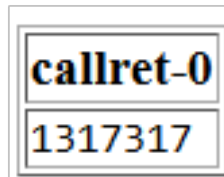


Figure 6: Total number of properties excluding `rdf:type` for the instances of `Politician` class.

If we compute the difference between the total number of properties found in query 1 and this one, we get that there are $1672243 - 1317317 = 354926$ `rdf:type` triples associated to instances of the `Politician` class.

If we only want to obtain each different property once, while still excluding `rdf:type`, we can combine `DISTINCT` with the same `FILTER`.

```

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT DISTINCT ?property
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .

    # Obtain every property used to describe those instances
    ?politician ?property ?value .

    # Exclude rdf:type
    FILTER (?property != rdf:type)

}

```

If we wanted, we can also count how many different properties there are, excluding `rdf:type`, by using `COUNT(DISTINCT ?property)`. This could be done just to make sure we've filtered just that one property, because the expected result should be one unit less than the one computed in the prior task ($1362 - 1 = 1361$).

```

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT COUNT(DISTINCT ?property)
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .
}

```

property
http://www.w3.org/2000/01/rdf-schema#label
http://www.w3.org/2000/01/rdf-schema#comment
http://www.w3.org/2002/07/owl#sameAs
http://xmlns.com/foaf/0.1/name
http://xmlns.com/foaf/0.1/topic
http://www.w3.org/2007/05/powder-s#describedby
http://xmlns.com/foaf/0.1/homepage
http://purl.org/dc/terms/subject
http://xmlns.com/foaf/0.1/isPrimaryTopicOf
http://es.dbpedia.org/property/gabinete
http://es.dbpedia.org/property/afiliaciones
http://es.dbpedia.org/property/almaMáter
http://es.dbpedia.org/property/almamáter
http://es.dbpedia.org/property/cargo
http://es.dbpedia.org/property/assessor

Figure 7: Different properties used by instances of Politician class, excluding `rdf:type`.

```
# Obtain every property used to describe those instances
?politician ?property ?value .

# Exclude rdf:type
FILTER (?property != rdf:type)

}
```

callret-0
1361

Figure 8: Number of different properties, excluding `rdf:type`, for the instances of Politician class.

3 Query 3

Natural language query

Which different values exist for the properties, except for `rdf:type`, applicable to the instances of `Politician`?

For this query, we will obtain the different values associated with the properties seen in the last task. Now the variable we want to return is `value` (instead of `property`). And since the question asks for different values, we use `DISTINCT` to avoid repeated results.

SPARQL query

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT DISTINCT ?value
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .

    # Obtain the properties and their corresponding values
    ?politician ?property ?value .

    # Exclude rdf:type
    FILTER (?property != rdf:type)
}
```

"Adame Ba Konaré"@es
"Adriano Sánchez Roa"@es
"Adrián Ward"@es
"Agustín Haya de la Torre de la Rosa"@es
"Agustín Molina Martínez"@es
"Ahmed Hilmi Pasha"@es
"Aileen Baviera"@es
"Aisha Musa el-Said"@es
"Akua Asabea Ayisi"@es
"Alain Vivien"@es
"Alaiza Pashkevich"@es
"Aptripel Tumimomor"@es
"Arancha González Laya"@es
"Archer Blood"@es
"Ariela Luna"@es
"Arnulfo Gasca Trujillo"@es
"Arturo Zúñiga"@es
"Ascensión Pastor Parres"@es
"Ashok-Alexander Sridharan"@es
"Aslan Bzhania"@es
"Adame Ba Konaré o Adame Konaré Ba (Segú, 1 de mayo de 1947) es una historiadora, escritora, militante de
"Adriano de Jesús Sánchez Roa (nacido el 24 de enero del año 1956) es un político, economista y escritor.
"Manuel Mariano Adrián Ward Chocano fue un hacendado, periodista y político peruano. Fue senador por Tacn
"Agustín Haya de la Torre de la Rosa (Lima, Perú, 3 de diciembre de 1948) es un sociólogo peruano."@es
"Agustín Molina Martínez (Distrito de Capaya, Provincia de Aymaraes, Apurímac, 27 de marzo de 1953), es u
"Ahmed Hilmi Abd al-Baqi (en árabe: أحمد حلمي عبد الباقي; nacido en 1883, Sidón, Imperio otomano- febrero de 19
"Aileen San Pablo, vda. de Baviera (Manila, 26 de agosto de 1959 - Ibidem., 21 de marzo de 2020) fue una
"Aisha Musa el-Said (también Asha, Ayesha, Mousa, Elsaid, El Said, Saeed, en árabe, عائشة موسى السعيد) es una
"Akua Asabea Ayisi (Mampong Akuapem, 3 de abril de 1927 - 21 de abril de 2010) fue una feminista, ex juez
"Alain Vivien (Malin, Calvados-Marna, 30 de agosto de 1938) es un político francés, miembro del Partido C

Figure 9: Different values associated with properties of instances of the `Politician` class, excluding `rdf:type`.

Since only `?value` is included in the `SELECT` clause, the result does not indicate which property each value belongs to. However we can include both to see how they are related (following task).

At this point, we only know that there are 438083 different values associated with the properties of instances of the `Politician` class. This implies that different properties may share the same value, but in this query we only consider one entry for each distinct value, regardless of which property it comes from.

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT (COUNT(DISTINCT ?value) AS ?numberOfDistinctValues)
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .

    # Obtain the properties and their corresponding values
    ?politician ?property ?value .

    # Exclude rdf:type
    FILTER (?property != rdf:type)
}
```

callret-0
438083

Figure 10: Number of different values associated with properties of instances of the `Politician` class, excluding `rdf:type`.

4 Query 4

Natural language query

For each of these applicable properties, except for `rdf:type`, which different values do they take globally for all those instances?

In the previous task, we obtained 438083 different values associated with properties of instances of the `Politician` class (excluding `rdf:type`). However, because only `?value` was returned, we did not know which property each value belonged to.

In this query, we return both `?property` and `?value`. This allows us to see which different values are associated with each property (except `rdf:type`).

The use of `DISTINCT` now applies to the complete pair (`?property`, `?value`). Therefore, the same property may appear several times with different values, and the same value may also appear several times if it belongs to different properties. What is not going to be repeated is the exact same property-value pair.

SPARQL query

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT DISTINCT ?property ?value
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .

    # Obtain the properties and their corresponding values
    ?politician ?property ?value .

    # Exclude rdf:type
    FILTER (?property != rdf:type)
}
```

property	
http://www.w3.org/2000/01/rdf-schema#label	"Adame Ba Konaré"@es
http://www.w3.org/2000/01/rdf-schema#label	"Adriano Sánchez Roa"@es
http://www.w3.org/2000/01/rdf-schema#label	"Adrián Ward"@es
http://www.w3.org/2000/01/rdf-schema#label	"Agustín Haya de la Torre de la Rosa"@es
http://www.w3.org/2000/01/rdf-schema#label	"Agustín Molina Martínez"@es
http://www.w3.org/2000/01/rdf-schema#label	"Ahmed Hilmi Pasha"@es
http://www.w3.org/2000/01/rdf-schema#label	"Aileen Baviera"@es
http://www.w3.org/2000/01/rdf-schema#label	"Aisha Musa el-Seid"@es
http://www.w3.org/2000/01/rdf-schema#label	"Akua Asabea Ayisi"@es
http://www.w3.org/2000/01/rdf-schema#label	"Alain Vivien"@es
http://www.w3.org/2000/01/rdf-schema#label	"Alaiza Pashkevich"@es
http://www.w3.org/2000/01/rdf-schema#label	"Aptripel Tumimomor"@es
http://www.w3.org/2000/01/rdf-schema#label	"Arancha González Laya"@es
http://www.w3.org/2000/01/rdf-schema#label	"Archer Blood"@es
http://www.w3.org/2000/01/rdf-schema#label	"Ariela Luna"@es
http://www.w3.org/2000/01/rdf-schema#label	"Arnulfo Gasca Trujillo"@es
http://www.w3.org/2000/01/rdf-schema#label	"Arturo Zúñiga"@es
http://www.w3.org/2000/01/rdf-schema#label	"Ascensión Pastor Parres"@es
http://www.w3.org/2000/01/rdf-schema#label	"Ashok-Alexander Sridharan"@es
http://www.w3.org/2000/01/rdf-schema#label	"Aslan Bzhania"@es
http://www.w3.org/2000/01/rdf-schema#comment	"Adame Ba Konaré o Adame Konaré Ba (Segú, 1 de mayo de 1947) es una histo
http://www.w3.org/2000/01/rdf-schema#comment	"Adriano de Jesús Sánchez Roa (nacido el 24 de enero del año 1956) es un
http://www.w3.org/2000/01/rdf-schema#comment	"Adrián Ward es un periodista, escritor y actor

Figure 11: Different values associated with each property of instances of the `Politician` class, excluding `rdf:type`.

For this reason, the number of results obtained with this query can be different (more) from the number obtained in Query 3. In Query 3, if the same value was associated with several different

properties, it was counted only once because only `?value` was considered.

In Query 4, however, the result is based on distinct `(?property, ?value)` pairs. Therefore, the same value can contribute to several different results if it is associated with different properties.

To count the total number of distinct property-value pairs returned by this query, we can use:

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT COUNT(*)
WHERE {
  {
    SELECT DISTINCT ?property ?value
    WHERE {
      # Find resources that are instances of Politician
      ?politician a dbo:Politician .

      # Obtain each property and its corresponding value
      ?politician ?property ?value .

      # Exclude rdf:type
      FILTER (?property != rdf:type)
    }
  }
}
```

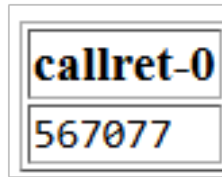


Figure 12: Number of distinct property-value pairs for instances of the `Politician` class, excluding `rdf:type`.

5 Query 5

Natural language query

For each of these applicable properties, except for `rdf:type`, how many distinct values do they take globally for all those instances?

In the previous task, we obtained the different `(?property, ?value)` pairs associated with instances of the `Politician` class. Now, instead of listing all those different values, we want to know how many distinct values are associated with each property. To do this, we group the results by `?property` using `GROUP BY`, and then count the different values in each group using `COUNT(DISTINCT ?value)`.

SPARQL query

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT ?property COUNT(DISTINCT ?value)
WHERE {
    # Find resources that are instances of Politician
    ?politician a dbo:Politician .

    # Obtain each property and its corresponding value
    ?politician ?property ?value .

    # Exclude rdf:type
    FILTER (?property != rdf:type)
}
GROUP BY ?property
```

property	callret-1
http://es.dbpedia.org/property/pieFoto	1
http://es.dbpedia.org/property/almamáter	2142
http://es.dbpedia.org/property/lugarMuerte	9
http://es.dbpedia.org/property/sitioInternet	1
http://es.dbpedia.org/property/fortuna	44
http://es.dbpedia.org/property/cupación	2
http://es.dbpedia.org/property/ideología	5
http://es.dbpedia.org/property/líder	1
http://dbpedia.org/property/knownFor	2
http://es.dbpedia.org/property/wikidata	15
http://es.dbpedia.org/property/acusaciones	3
http://es.dbpedia.org/property/fórmula	73
http://dbpedia.org/ontology/president	3
http://es.dbpedia.org/property/ciudadanía	15

Figure 13: Number of distinct values associated with each property of instances of the `Politician` class, excluding `rdf:type`.